

Business Requirements Document

HR Attrition Analytics Dashboard (Power BI)

Document control	
Project	HR Attrition Analytics Dashboard
Version / status	1.0 – Draft
Date	21 September 2026
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Sponsor (simulated)	Head of People and Culture
Data source	IBM HR Employee Attrition dataset (synthetic; 1,470 employees, 35 fields)

1. Purpose and background

The People and Culture team has seen rising staff turnover: 237 of 1,470 employees have left, an overall attrition rate of 16.1%. Leadership does not have a consistent view of where attrition is happening, which groups are most affected, or what is driving it.

This project delivers an interactive Power BI dashboard, built on a governed data model and standard KPI definitions, so HR can prioritise retention actions using evidence. The scenario is fictional and the data is synthetic; the document follows the structure of a real business requirements specification.

2. Business objectives

- Provide HR with a single, trusted view of workforce and attrition KPIs.
- Identify the departments, roles, and employee groups with the highest attrition.
- Surface the main drivers of attrition (workload, pay, tenure, satisfaction) so interventions can be targeted.
- Estimate the potential impact of addressing the largest driver.

3. Scope

3.1 In scope

- A three-page Power BI report covering overview, drivers, and talent and risk.
- A star schema data model with documented, explicit measures.
- Written findings and recommendations, with limitations stated.

3.2 Out of scope

- Real-time or scheduled data refresh, and row-level security.

- Cost-of-attrition modelling.
- Trends over time, because the dataset contains no date field.
- Predictive risk scoring (a stretch goal, see BR-09).

4. Stakeholders (simulated)

Stakeholder	Interest
Head of People (sponsor)	Overall attrition picture and where to focus retention effort.
HR Business Partners	Department- and role-level detail to support managers.
Compensation and Benefits	Relationship between pay, stock options, and attrition.
Line managers	Team-level risk factors such as workload and travel.

5. Business requirements

Priority uses MoSCoW: Must, Should, Could, Won't (this release).

ID	Requirement	Priority	Delivered on
BR-01	View headline KPIs (headcount, leavers, attrition rate, average tenure, average income), filterable by department, gender, and job level.	Must	Page 1
BR-02	See where attrition concentrates by department, job role, age group, and tenure, showing both rate and volume.	Must	Page 1
BR-03	Compare every attrition rate against the overall attrition rate.	Must	Pages 1–2
BR-04	Analyse drivers: overtime, business travel, marital status, income band, stock options, and satisfaction measures.	Must	Page 2
BR-05	Identify high-risk employee segments, with group size and attrition rate.	Should	Page 3
BR-06	Assess whether high performers and highly involved staff are leaving.	Should	Page 3
BR-07	Quantify the leavers that could be avoided if overtime-related attrition matched the baseline rate.	Should	Page 3
BR-08	Provide written findings and recommendations in plain language.	Must	Page 3 and README
BR-09	Score individual employees by attrition risk using a predictive model.	Could	Future release
BR-10	Show attrition trends over time.	Won't	No date field in source

6. Data requirements

6.1 Source and preparation

- **Source:** IBM HR Employee Attrition dataset, 1,470 rows and 35 columns, no null values, unique EmployeeNumber.
- **Cleaning:** remove the constant columns EmployeeCount, StandardHours, and Over18; derive a 1/0 IsAttrition field from Attrition.
- **Labelling:** replace the numeric 1–4 and 1–5 rating scales (satisfaction, involvement, work-life balance, education) with descriptive labels.

6.2 Data model

A star schema with one fact table at employee grain and the following dimensions:

- DimDepartmentRole: 11 department and job role combinations. Some roles span departments, so the key is the combination.
- DimEducationField: education field and education level labels.
- DimEmployeeProfile: gender, marital status, business travel, overtime, and stock option level.
- DimAgeGroup and DimSalarySlab: editable band definitions.
- DimRating: labels for the 1–4 rating scales.

6.3 KPI definitions

KPI	Definition	Source fields
Headcount	Number of employees in the current filter context.	Row count of FactEmployee
Attrition Count	Number of employees who have left.	IsAttrition
Attrition Rate	Attrition Count ÷ Headcount.	IsAttrition
Overall Attrition Rate	Attrition Rate with all filters removed; used as the comparison baseline.	IsAttrition
Rate vs Overall (pts)	Attrition Rate minus Overall Attrition Rate, in percentage points.	Derived
Avg Monthly Income	Average of monthly income.	MonthlyIncome
Avg Tenure (Yrs)	Average years at the company.	YearsAtCompany

7. Non-functional requirements

- **Usability:** each page answers one business question, with takeaway-style titles and consistent colour use.
- **Reliability:** all visuals use explicit measures, and totals reconcile to the source data.
- **Privacy and ethics:** the data is synthetic. In a real deployment, apply a minimum group size (for example, flag or hide groups with fewer than 10 employees) and restrict access to sensitive fields such as gender and marital status.
- **Accessibility:** do not rely on colour alone to convey meaning, and keep text legible.

- **Documentation:** a README containing the model diagram, measure list, findings, and limitations.

8. Assumptions, constraints, and risks

Assumption / risk	Mitigation
The data is synthetic and imbalanced (about 16% leavers), so findings demonstrate method rather than real-world truth.	State this in the README and on the report.
Some segments are small (for example, HR and Sales Representatives), which risks overinterpreting their rates.	Show group size (n) beside every rate.
Correlation is not causation: overtime overlaps with role, level, and income.	Present the overtime scenario (BR-07) as an illustration, not a forecast.
There is no date field, so trends cannot be analysed.	Optionally derive a synthetic hire date from YearsAtCompany and label it clearly as constructed.

9. Acceptance criteria

- Headcount is 1,470, leavers are 237, and the overall attrition rate is 16.1%, matching the source data.
- Every rate visual shows group size, either on the chart or in the tooltip.
- Every slicer cross-filters all visuals on its page.
- Each business requirement is traceable to at least one visual (traceability table in the README).
- Findings, recommendations, and limitations are written in plain language.

10. Sign-off

Role	Name	Date
Business sponsor		
Business analyst		
Report developer		